

Conclusion to Python Fundamentals

 <https://github.com/learn-co-curriculum/phase-3-conclusion-to-python-fundamentals>  <https://github.com/learn-co-curriculum/phase-3-conclusion-to-python-fundamentals/issues/new>

Learning Goals

- Understand how to learn a new programming language.
 - Learn the benefits and philosophy of Python as a language.
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Key Vocab

- **Interpreter:** a program that executes other programs. Python programs require the Python interpreter to be installed on your computer so that they can be run.
 - **Python Shell:** an interactive interpreter that can be accessed from the command line.
 - **Data Type:** a specific kind of data. The Python interpreter uses these types to determine which actions can be performed on different data items.
 - **Exception:** a type of error that can be predicted and handled without causing a program to crash.
 - **Code Block:** a collection of code that is interpreted together. Python groups code blocks by indentation level.
 - **Function:** a named code block that performs a sequence of actions when it is called.
 - **Scope:** the area in your program where a specific variable can be called.
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Conclusion

You've learned a lot in this first module of Phase 3:

- How to execute code using the Python shell.

- Python's many data types.
- Common errors and Exceptions in Python.
- How Python handles scope and organizes code blocks using indentation.
- How to control flow using functions, conditionals, and loops.

Python shares many similarities with JavaScript, but it presents many added benefits as well. Your new knowledge of the fundamentals of Python programming will allow you to create powerful applications that work closely with the data on your computer (as well as other servers and databases!) to accomplish almost any task you can imagine.

Resources

- [About Python](https://www.python.org/doc/essays/blurb/) ➞ (https://www.python.org/doc/essays/blurb/)
- [Python documentation](https://docs.python.org/3/) ➞ (https://docs.python.org/3/)
- [Python Style Guide](https://peps.python.org/pep-0008/) ➞ (https://peps.python.org/pep-0008/)
- [Python Data Types](https://docs.python.org/3/library/datatypes.html) ➞ (https://docs.python.org/3/library/datatypes.html)
- [Python Scope of Variables - GeeksforGeeks](https://www.geeksforgeeks.org/python-scope-of-variables/) ➞ (https://www.geeksforgeeks.org/python-scope-of-variables/)
- [Control Flow Statements - O'Reilly](https://www.oreilly.com/library/view/python-in-a/0596001886/ch04s09.html) ➞ (https://www.oreilly.com/library/view/python-in-a/0596001886/ch04s09.html)
- [Guido van Rossum: The Early Years of Python - computer.org](https://www.computer.org/csdl/magazine/co/2015/02/mco2015020007/13rUy3gmYB) ➞ (https://www.computer.org/csdl/magazine/co/2015/02/mco2015020007/13rUy3gmYB)